edikit: A Python Toolkit for Reproducible Construction and Auditing of National Social Accounting Matrices

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Abstract

`edikit` is a Python library for building and auditing national social accounting matrices (SAMs) from official statistics. It brings established methods (supply-use reconciliation, concordances, and RAS and GRAS balancing) into one documented workflow that reports every change it makes to the data. At each stage the library computes and reports the accounting identities, and it classifies accounts only through explicit mappings. It returns every balancing adjustment cell by cell, reports and marks the entry that closes an account, and can verify registered inputs against SHA-256 manifests. The evidence comes from the package's scripted workflows, or "recipes." One builds balanced macro SAMs from Eurostat's harmonized data. It is shown for Belgium, Denmark, and France; for Spain, where it flags government accounts missing from current data; and in a coverage test across all 37 candidate countries. Another audits a distributed SAM against published controls, shown for the 2019 SAM for Ghana. A third reproduces a separately reported cell-level reconstruction of the 2019 South African SAM. The examples show that the software runs as documented, that its balanced outputs are accounting-consistent, and that it reports when balancing fails, as for Spain. They do not establish a general procedure for constructing a national SAM. Outside the Eurostat datasets, the library supplies the components for building one from national supply and use tables (readers, concordances, explicit classification, and balancing), while the construction rules remain country-specific and require source interpretation and economic judgment. The package has one required dependency, `openpyxl`, and is installable from PyPI.

Keywords: social accounting matrix; supply and use tables; national accounts; reproducibility; RAS balancing; GRAS; data provenance; Eurostat; Python.

1 Overview

1.1 Introduction

A social accounting matrix (SAM) is a square table recording, for one economy and one year, every payment made from one account to another (1). Rows and columns carry the same ordered list of accounts: activities that produce, commodities that are produced, factors of production such as labor and capital, and institutions such as households, corporations, and government, together with capital accumulation and the rest of the world. The cell in row i,

column j holds the payment from account j to account i, so a row gives an account's receipts and a column its outlays. For every account, the two totals must agree, as every payment is someone else's income. National SAMs range from a few dozen accounts to several hundred and follow the conventions of the System of National Accounts (2).

The matrix is a database, not a model, but much applied economy-wide policy analysis rests on it (3). Multiplier analysis traces how a change in demand propagates through the structure it records, and computable general equilibrium models are calibrated so that they reproduce the SAM in their base year before any policy is simulated. Ministries of finance, central banks, development banks, and research institutes use such models to assess tax reforms, tariff changes, and public-investment programs. A model inherits whatever the matrix contains, so an error in a cell carries through, usually unnoticed, into results read as policy advice.

Building a SAM is a data-reconciliation exercise (4). The inputs are official statistics: supply and use tables (SUTs),[1] national and sector accounts, balance-of-payments statistics, and household and labor force surveys. Each source has its own classification, units, and vintage, and covers only part of what the matrix needs. The analyst who builds the matrix assigns each source category to a SAM account, estimates the cells that no source reports, and chooses between sources that give different figures for the same flow. The assembled matrix then does not balance, because independent sources do not satisfy the identity by construction, and a final step adjusts cells until receipts equal payments everywhere. This is usually done by the biproportional method known as RAS (the name is not an abbreviation) or by cross-entropy estimation (6). Even an unusually well-documented SAM, such as the 2019 South African matrix of van Seventer and Davies (7), comes as a matrix with a technical note and no code to regenerate it from its sources.

`edikit` was written for that problem, during a cell-level replication audit of the van Seventer and Davies matrix, published by the United Nations University World Institute for Development Economics Research (UNU-WIDER) and reported separately (8). This paper describes the software itself. It sets out which properties of a matrix the software guarantees and which it leaves to the analyst, and it presents evidence that the software works as documented. None of the worked examples reported here appears in that earlier study.

`edikit` adds no new SAM mathematics. RAS, generalized RAS (GRAS) for matrices with negative entries (9, 10), cross-entropy balancing, and the accounting identities are long established, and tools for them already exist. They include published General Algebraic Modeling System (GAMS) programs for cross-entropy SAM estimation (11), extensions of RAS for conflicting information (12) and for joint supply-use balancing (13), collaborative laboratories that compile and balance input–output systems at scale (14), the compilation methodology behind Eurostat's inter-country supply-use tables (15), and general-purpose fitting packages such as `mipfp` (16). Two are especially close to this work. The European Commission's Joint Research Centre builds SAMs for EU member states from Eurostat's supply-use tables and

[1] Supply and use tables are the pair of matrices national statistical offices publish to describe production and its disposition. The supply table records which industries produce which products; the use table records which industries and which final users consume them, together with the value added generated in production (5). Together they satisfy the identities that anchor a SAM's production block: the total supply of each product equals its total use, and each industry's output equals its intermediate consumption plus value added.

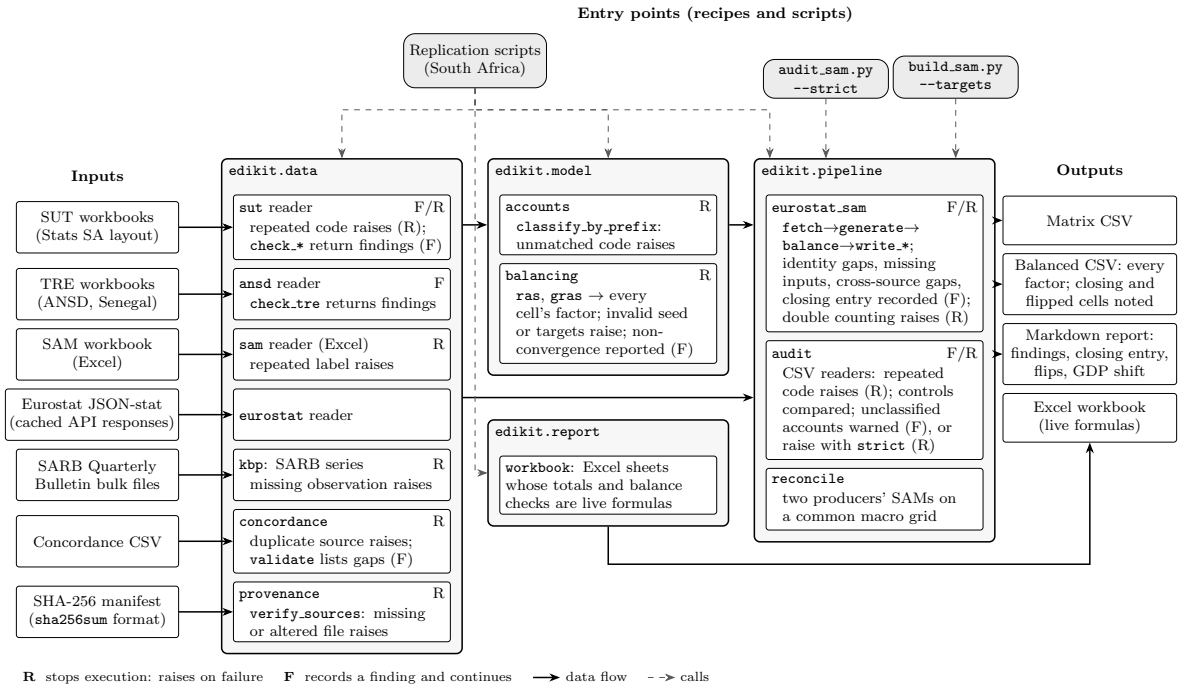


Figure 1: Architecture of `edikit`: packages, inputs and outputs, entry points, and how each check handles failure (R: raises and stops; F: records a finding and continues). Components without a mark neither stop execution nor record findings. Every module can also be called directly from user code.

sector accounts (17), and the R package `iotables` retrieves and processes Eurostat's symmetric input–output tables (18). Further downstream, `pymrio` analyzes compiled multi-regional input–output databases (19), and `MARIO` builds and analyzes input–output models from supply-use and input–output tables (20). The scope of `edikit` is narrower. It keeps a matrix's whole path, from registered source files to published output, inside one open Python library: it reads the sources, reports which accounting identities fail, maps accounts only by explicit rule, logs every balancing adjustment, and verifies that the inputs are the registered ones. It is not a general SAM-construction system, and it does not implement cross-entropy estimation.

1.2 Implementation and architecture

`edikit` is pure Python (about 1,630 statements) with one required external dependency, `openpyxl`, for Excel input and output. It has four packages. Figure 1 shows the packages, the inputs each reader accepts, the outputs the pipelines write, the entry points that call them, and how each check handles failure.

- `edikit.data`: functions that read official input files. Each reader handles one publisher's file format. They cover SUT workbooks in the layout of Statistics South Africa; supply-use tables (*tableaux des ressources et des emplois*, TRE) in the layout of Senegal's Agence Nationale de la Statistique et de la Démographie (ANSD); SAM workbooks in Excel; Eurostat JSON-stat responses; and the bulk series files of the South African Reserve Bank (SARB), with an evaluator for signed series formulas. The package also holds

Stops execution (raises)	
`verify_sources`	a registered file is missing or its digest differs from the manifest; a manifest line is malformed or repeated
`classify_by_prefix`	an account code matches no declared rule
`ras`, `gras`	row and column targets disagree in total; for RAS, a negative seed or a positive target on an empty row or column
`concordance`	a source code is mapped twice; aggregating a code with no mapping
readers	a repeated cell, row code, column code, or label (SUT, SAM-workbook, and audit CSV readers; classification and controls files); an unknown account kind
`audit(strict=True)`	an unclassified account, or a classification for an account the matrix lacks
`eurostat_sam.generate`	no data for the country-year; leaf industries exceeding the published total by more than 5% (double counting); an industry absent from the use table
Recorded, execution continues	
SUT and ANSD `check_*`	identity violations
`concordance.validate`	unmapped or unknown source codes
`eurostat_sam.generate`	identity findings; supply-use cross-differences; suppressed detail; sector-account items published on neither side; differences between the supply-use tables and the sector accounts; the closing entry
`eurostat_sam.balance`	negative cells flipped before RAS
`audit` (default)	unclassified accounts (left out of aggregates) and classifications for absent accounts; accounts whose imbalance exceeds what rounding can produce; deviations from controls
`ras`, `gras`	non-convergence (`converged=False`)

Table 1: The checks: which stop execution and which are recorded. A finding is returned to the caller and written to the report; whether to proceed is the user's decision. Parsing failures, such as a missing sheet, block, or series observation, also raise.

concordances and `provenance`, which verifies files against a SHA-256 manifest. A source in another format needs a new reader.

- `edikit.model`: the list of account types (activity, commodity, factor, household, and so on) and RAS and GRAS balancing. Each account's type comes from a table of rules the user writes, never from guessing at its label. `classify_by_prefix` applies the table and stops with an error on any account that no rule covers. The balancing routines return the balanced matrix together with the factor by which each cell was scaled.

- `edikit.pipeline`: `eurostat_sam` generates, balances, and reports a macro SAM for country-years covered by the European System of Accounts (ESA 2010) transmission datasets (21). `audit` reads a distributed SAM in CSV, checks its structure, and compares its macro aggregates with published national-accounts controls. `reconcile` places two producers' SAMs of the same economy on a common macro grid.

- `edikit.report`: Excel workbooks whose totals and balance checks are live formulas instead of pasted values, so a reader can trace them in a spreadsheet program.

Table 1 sorts the checks into two groups: those that stop the run and those that record a finding and let it continue. The split is deliberate. A published table can fail an accounting

identity simply because the publisher has withheld confidential cells. Stopping on such a failure would make the table unusable, so the failure is recorded instead, which documents it. Problems that would silently corrupt a result, such as an identifier that appears twice or industries counted twice, stop the run. The double-counting check allows a 5 percent margin because a source's own detail can exceed its published total, at the cost of missing a double-counted section smaller than that. Parsing and checking are kept apart: the readers only parse, the `check_*` functions then test the parsed table against its identities, and a pipeline writes its report when its `write_report` is called, as the recipes do. Every check either raises an exception or returns a finding, and none relies on `assert`, so running Python with `-O` removes none of them.

Provenance is recorded partly by hand. A `SOURCES.md` file states what each source is, who publishes it, how it was obtained, and under what license. Whether a local file is the registered one is checked automatically by `verify_sources`, which compares each file with a manifest in `sha256sum` format and raises on a missing or altered file. `write_manifest` generates such a manifest, and standard tools (`shasum -a 256 -c`) read the same file. A script must call this verification explicitly, since neither `fetch` nor the recipes do.

The Eurostat pipeline embeds four economic choices.

Instrument accounts. Distributive transactions (property income, current taxes on income and wealth, social contributions and benefits, other current transfers, pension-entitlement adjustments, and capital transfers) are routed through instrument accounts: each institutional sector pays into an instrument account and receives from it. The sector accounts record what each sector pays and receives but not to whom, so the routing preserves every sector's totals without inventing bilateral flows. A model that uses the matrix will nonetheless distribute a pooled account's receipts in fixed shares, which amounts to assuming proportional from-whom-to-whom flows.

Account residuals and their structure. Before balancing, the institutional accounts carry residuals. The report sets beside each residual the items computed from the inputs that bear on it, together with what they leave unexplained. Some items are cross-source differences: the supply-use tables and the sector accounts each measure gross value added, compensation of employees, operating surplus, and household consumption, and the report shows both measurements with each dataset's vintage. Others follow from the account design. Net capital transfers stay in the institutions' accounts, because the savings–investment account receives saving before capital transfers. For EU members, part of the taxes on production and imports, mainly customs duties, is paid to EU institutions, which the sector accounts treat as part of the rest of the world. The generator credits all such taxes, net of the subsidies the rest of the world pays, to the domestic government, so the government account carries the difference. Purchases less sales of non-produced assets, such as land and licenses, are not represented, so the savings–investment account carries their net value. The examples show how far these items go. In all four generation examples (Table 2), the residual of the labor account equals the difference between the two datasets' figures for compensation of employees, and in three of the four, the residual of the corporations' account equals their net capital transfers. When Eurostat publishes a sector-account item neither as paid nor as received, the generator sets it to zero and lists it, with a warning that any account it feeds is unreliable.

The closing entry. The capital factor account is closed by a residual. Gross operating surplus and mixed income enter it from the supply-use tables, the sector accounts distribute their own total to domestic sectors, and the difference is booked against the rest of the world. This closing entry measures the gap between the two datasets' operating-surplus figures and records no observed flow. Because it is booked against the rest of the world, it also appears in that account's residual. Every report states it, and the balanced CSV marks it.

Balancing. Each account needs one target. The default is the mean of its receipts and payments, which treats the two sides as equally reliable; receipts alone or payments alone can be chosen instead with the `--targets` option. Before RAS, a negative cell is replaced by a positive flow in the opposite direction, which keeps each account's net position but changes its gross totals. A negative closing entry therefore reappears after balancing as a positive flow from the rest of the world to the capital factor account, where it reads like factor income from abroad; the balanced CSV's `note` column marks such cells. The library's GRAS keeps negative entries in place; the pipeline uses the flip and RAS instead. Balancing also changes aggregates the sources publish consistently: gross domestic product (GDP) moves even where the supply-use block has no identity findings, and no cell or block can be held fixed.

Three documented *recipes* serve as entry points. `build/` generates and balances a macro SAM for a country-year (`python build_sam.py --country DK --year 2019`). `audit/` audits a distributed SAM in CSV (`python audit_sam.py --sam matrix.csv --kinds kinds.csv --controls controls.csv`). `replicate/` documents how the scripts of the South African reconstruction reproduce its reported numbers. The same steps take a few lines of Python:

```
from edikit.pipeline import eurostat_sam as es

paths = es.fetch("BE", 2019, "data/")     # three requests, cached
res   = es.generate("BE", 2019, paths)    # identities computed and recorded
bal, flipped = es.balance(res)            # RAS; every factor in bal.factors
es.write_report(res, "BE_2019_generation.md",
                balance=bal, flipped=flipped)
es.write_balanced_csv(bal, "BE_2019_macro_sam_balanced.csv",
                      flipped=flipped, plugs=res.plugs)
```

For the Belgian inputs distributed in the repository, the report reads in part (quoted verbatim, with long lines wrapped and one cut marked):

```
- Output and VA identities: 0 findings at EUR 1m tolerance
- GDP at basic prices (generated): EUR 427,315m
| Gross value added at basic prices | 427,315.1 | 429,084.0 |
  +1,768.9 | +0.41% |
- (row,cap) = EUR -1,220.7m (-0.286% of GDP)
- Negative cells flipped to their positive transpose before
  balancing: (atx,act), (gvt,atx), (row,cap), (d9,corp)
- GDP at basic prices after balancing: EUR 431,146.3m (+0.897%
  vs. generated); [...]
```

The value-added row compares the two datasets' measurements, the (row,cap) line is the closing entry, the list of flipped cells shows it again, and the last line shows that balancing moves GDP.

1.3 Quality control

Automated tests. The public test suite has 78 tests. Many of them use accounting identities as the check: they test whether the readers, the account classification, RAS and GRAS, and the audit produce results in which, for example, supply equals use and each account's receipts equal its payments. Other tests run the whole generation and balancing pipeline on a small invented economy with two industries, built so that every identity holds exactly, so any departure signals an error. Each check in Table 1 is exercised by at least one test, including every type of finding the identity checks record, as are the residual diagnostics, the CSV notes, the rounding test, the alternative balancing targets, and the output precision. Line coverage is 95 percent (90–99 percent per module). Coverage shows only which code paths run, so the evidence of accounting correctness comes from the identity-based and end-to-end tests. Continuous integration runs the suite on Linux, macOS, and Windows with Python 3.11–3.13 for every pull request and every push to the main branch. The wheel installed by `pip install edikit` contains the library only; the tests are included in the source distribution and the repository.

Worked examples. Every input the worked examples need is distributed in the repository and registered in SHA-256 manifests, so each example runs offline and reproduces the outputs stored with it. Table 2 summarizes them together with the coverage test, which fetches current data. They cover two tasks: generation within a harmonized statistical system, and the audit of a matrix someone else built. They test execution and accounting consistency and leave open whether the construction rules produce the economically preferred matrix.

The generation inputs are Eurostat's supply table, use table, and non-financial sector accounts (22, 23, 24). Eurostat revises them, so the same command run later can give different results, and every report records the vintage of each dataset. In the October data, seven countries (Austria, Germany, Hungary, Italy, Portugal, Spain, and Sweden) lack nine general-government sector-account items, and their reports name them. A country passes when it has zero identity findings, full commodity closure, no supply-use cross-differences, an import identity within tolerance, no suppressed detail, no unpublished sector-account items, converged balancing, no residual above 5 percent of GDP, and no factor more than 100 percent away from one. Passing does not mean balancing changed little: France passes although one of its cells was scaled by 0.68, and several countries that need review have residuals above 10 percent of GDP.

The balancing target matters at the aggregate level. For Belgium, GDP after balancing moves by +0.86, +0.90, and +0.94 percent with receipts, mean, and payments targets, respectively. The matrix CSVs round values to EUR 0.1 million and factors to six decimals, so a balanced matrix closes to the solver's tolerance in memory but only to that rounding in the file; `--decimals` writes a higher-precision export. The toolkit's text reports are written with LF line endings on every platform, so the reports, like the CSVs, are byte-identical across operating systems.

For Ghana, the audit uses the SAM published by the Ghana Statistical Service, the Institute of Statistical, Social and Economic Research, and the International Food Policy Research Institute (IFPRI) (25, 26), in GHS million, with controls from the World Bank's World Development Indicators (27). The audit locates departures from the controls but cannot explain

Example (2019)	Task	What the run reports
Belgium	generation and balancing	0 identity findings; GVA gap +0.41%; largest residual 0.73% of GDP; closing entry −0.29% of GDP; factors in [0.915, 1.063]; GDP +0.90% after balancing
Denmark	generation and balancing	0 identity findings; the datasets agree on GVA, compensation, and operating surplus (closing entry EUR 1 million) but differ by −1.11% on household consumption; largest residual 0.20% of GDP; factors in [0.988, 1.009]; GDP unchanged
France	generation and balancing	0 identity findings; GVA gap −0.86%; largest residual 2.07% of GDP; closing entry +1.56%; factors in [0.680, 1.095]; GDP −0.02%
Spain	generation from a release without government accounts	nine general-government sector-account items unpublished and listed; social-benefits residual −17.3% of GDP; balancing does not converge
37 countries	coverage test (October 2026 data)	6 pass every core check; 12 generate and balance with caveats; 10 need review; 2 cannot be balanced; 7 not covered
Ghana	audit of the published SAM against World Development Indicators controls	every account classified (`strict` passes); balanced to four decimals; production side within 0.1% of controls; private consumption −2.2%, exports +1.9%, changes in inventories +167% on a small base

Table 2: Worked examples (shipped with their inputs and runnable offline) and the coverage test (run against the live API), by task. Generation examples use October 2026 Eurostat data. GVA: gross value added. All figures are taken from the reports and coverage table the recipes write.

them. The aggregates also depend on the account classification and on the audit's definitions. Private consumption, for example, includes households' purchases from activities as well as from commodities. For matrices distributed in rounded values, the audit tests each account's imbalance against the most rounding could produce: half the rounding unit (the grain) for each nonzero off-diagonal cell in the account's row and column. The bound omits cells rounded to zero, so it can understate the rounding error.

The South African reconstruction. The library was first used for a cell-level reconstruction of the 2019 South African SAM from public sources, informed by the benchmark it audits (8). That reconstruction reproduces the benchmark exactly (to its rounding unit of ZAR 0.5 million) in the production structure of all 61 activities, in the 316 final-demand cells, and in 43 of 45 computable macro cells, and agrees within ZAR 0.5 million on 62.5 percent of the 6,732 cells common to the two matrices. Where public sources do not reach the benchmark's household, occupation, and capital detail, it uses survey-based shares. Because the reconstruction is benchmark-informed, it bears on the benchmark's reproducibility and cannot independently validate `edikit`, so it is not counted here. Reproducing it requires third-party inputs that cannot be redistributed; the repository records how to obtain each one and the digest needed to verify each downloaded file.

2 Availability

2.1 Operating system

Pure Python; the test suite runs in continuous integration on Linux, macOS, and Windows.

2.2 Programming language

Python $\geq$ 3.11.

2.3 Additional system requirements

None; national SUT and sector-account inputs are a few megabytes.

2.4 Dependencies

`openpyxl`. Optional: `certifi`, used when installed to supply a TLS trust store for Eurostat API requests on platforms without system certificates; the `replicate` extra (`xlrd`, `matplotlib`) for two of the South African replication scripts; and the `dev` extra (`pytest`, `pytest-cov`) for the tests.

2.5 Software location

Package index: PyPI, `https://pypi.org/project/edikit/`; `pip install edikit` installs the library, and a release can be pinned (`pip install edikit==0.2.7`). The source distribution adds the tests and recipes.

Code repository: GitHub, `https://github.com/kodzovee9/tekmeris` (toolkit under `toolkit/`); license Apache-2.0; version 0.2.7, October 2026.

Archive: Zenodo, DOI 10.5281/zenodo.23723993 (version v0.2.7; concept DOI for all versions: 10.5281/zenodo.21419195); license Apache-2.0.

2.6 List of contributors

Kodzovi Abalo (design, development, validation).

2.7 Language

English.

3 Reuse potential

The package can be reused at three levels, set out below: parts that work for any economy, parts that work only within a particular statistical system such as Eurostat's, and work that cannot be automated because it requires the analyst's own economic judgment.

Parts that work for any economy. These need no change: the account types and rule-based classification, RAS and GRAS balancing with the adjustment factor of every cell, concordances

Recipe	Inputs	Outputs	Limits
`build/`	country code and year; Eurostat API, or cached responses (examples shipped)	matrix CSV; balanced CSV with every factor and notes; report	ESA-covered country-years only; results depend on the data vintage; macro SAM, not a disaggregated SAM
`audit/`	SAM in CSV; optional classification and controls (examples shipped)	report: balance, rounding test, aggregates against controls	CSV input only; reports differences, not their causes
`replicate/`	public inputs (in the repository) and restricted inputs (obtained by the user)	the South African reconstruction's numbers	1 of the 13 South African scripts runs from a clean clone; the rest need the restricted inputs

Table 3: The recipes: what each needs, what it produces, and where it stops.

that refuse duplicate mappings and report gaps, source verification, the Excel workbooks with live formulas, and the audit of any SAM saved as a CSV file. A modeler who uses a SAM built by someone else can audit it with two small files, one classifying its accounts and one giving the published national-accounts totals. The audit shows whether the matrix balances, whether its imbalances could come from rounding, and how its totals compare with the national accounts.

Parts tied to a statistical system. The Eurostat generator works only for country-years the ESA 2010 transmission datasets cover, and its results depend on the data vintage. The committed coverage table records which countries pass and which data it used. Each reader is tied to one publisher's layout.

Work that requires the analyst's judgment. Outside a harmonized system, building a SAM means writing readers for the national sources, mapping their classifications onto SAM accounts, choosing how to allocate flows the sources do not report, choosing balancing targets, and deciding how to treat conflicts between sources. The South African replication scripts work through these decisions for one country. They show the shape of the work but cannot simply be reconfigured for another country. `edikit` records such decisions and makes them checkable, and the analyst still has to make them.

Table 3 summarizes the recipes. Someone reading a report from the build recipe should work through it in four steps. First, check whether any sector-account figures are missing from Eurostat's data, because the generator sets them to zero and any account that uses them is unreliable. Second, read the table comparing the supply-use tables with the sector accounts, and the items listed beside each account's residual: together they show where the two datasets disagree and which residuals come from the way the accounts are set up. Third, look at the failed accounting identities; in a table with confidential cells withheld, these usually reflect the withholding rather than an error. Last, look at the balancing factors: a factor far from one marks a cell that balancing changed substantially, which should be examined before the matrix is used.

A balanced macro SAM is not a disaggregated SAM, and it may need further disaggregation by industry, household group, or factor before it suits a particular modeling application. Where several admissible mappings exist, an accounting-consistent output need not be the economically

preferred one; the toolkit shows the chosen mapping and every adjustment but does not certify them. Statistical confidentiality can prevent full reconstruction, because suppressed cells can be detected and quantified but not recovered. Balancing cannot hold reliable cells or blocks fixed or weight cells by reliability, as the KRAS extension of RAS does (12). Finally, `edikit` does not implement cross-entropy estimation against a prior (6), which matters most where margins are only partly known or further constraints apply.

Natural extensions include GRAS as a balancing option in the Eurostat pipeline, fixed and reliability-weighted cells, cross-entropy balancing behind the same diagnostic interface, multi-sector generation from the 65-industry ESA core, and survey-driven household disaggregation. User support, bug reports, and feature requests go through the repository's GitHub Issues page; contributions are welcome and must pass the test suite.

Data accessibility statement

The versioned software, the inputs of the worked examples, their generated outputs and reports, and the redistributable inputs of the South African reconstruction are archived on Zenodo at doi: 10.5281/zenodo.23723993. The inputs of the worked examples, including the audit classification and controls files, and the downloaded inputs of the South African reconstruction are registered in SHA-256 manifests named `SOURCES.sha256`, which `verify_sources` or the standard `shasum` tool can check. Inputs whose licenses do not permit redistribution are not included. For each, the reconstruction's `SOURCES.md` records the publisher, source location, access date, and license or access conditions, and its manifest records the digest needed to verify a locally obtained copy. The Eurostat inputs (22, 23, 24) are redistributed with attribution, as Eurostat permits. The Ghana example uses the 2019 SAM for Ghana (25, 26) and controls from the World Development Indicators (27), both under CC BY 4.0.

Generative AI use

Generative AI (Anthropic's Claude) was used for code generation and diagnosis. All generated code, mappings, and outputs were reviewed by the author and accepted only after the deterministic accounting checks and tests described above were satisfied. The author assumes full responsibility for the software.

Author roles

Kodzovi Abalo: conceptualization, software design and development, validation, writing.

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Economic Research, IFPRI, the World Bank, Statistics South Africa, the South African Reserve Bank, and UNU-WIDER.

Disclaimer

The views expressed here are those of the author and do not necessarily reflect the views of the World Bank, its Executive Directors, or the governments they represent.

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